PRODUCT REQUIREMENT DOCUMENT (PRD)

Project Zync

The Sovereign AI for Rare Earth Engineering

UMHackathon 2026 Domain:

AI for Economic Empowerment & Decision Intelligence

Team: Gunners

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Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysia holds an estimated 16.1 million tonnes of rare earth elements (REE) valued at approximately RM 1 trillion — one of the most strategically critical mineral reserves on the planet. Following the government's 2024 ban on raw REE exports under the 13th Malaysia Plan (13MP), Malaysia is legally and economically committed to processing these minerals domestically. However, the country faces a critical engineering intelligence gap: Malaysian site engineers and process managers must make high-stakes operational decisions — diagnosing yield failures, selecting leaching chemistry, and prioritising extraction zones — using fragmented geological data, manual calculations, and no intelligent decision support system. The only country that possesses the deep hydrometallurgical expertise to solve these problems at scale is China — the same technological monopoly Malaysia is strategically trying to reduce its dependence on. Every wrong decision costs millions in wasted reagents, lost yield, and regulatory penalties. Every month of delay keeps RM 1 trillion in the ground.
- **Target Domain:** AI-powered multi-agent decision intelligence for Malaysian rare earth element extraction operations — directly supporting the 13MP national economic agenda and Malaysia's ambition to become the world's primary source of heavy rare earth elements.
- **Proposed Solution Summary:** Zync is a six-agent AI decision intelligence system powered by Z.AI's GLM that gives Malaysian REE site engineers access to expert-level process reasoning that currently only exists inside closed Chinese state enterprises. The system routes operator inputs through a specialised pipeline: a GraphRAG historian agent retrieves relevant Malaysian geological precedents, a SciGLM chemist agent reasons through the optimal process chemistry with a visible step-by-step chain, a hybrid-search compliance agent blocks illegal configurations in real time, and a synthesis agent produces a clear, auditable recommendation with full reasoning trace. Remove GLM and the system cannot reason across chemical

variables, historical data, and regulatory constraints simultaneously — it becomes a static form. With GLM, Malaysia's engineers are augmented into midstream plant architects who can solve in minutes what previously took months of expert consultation.

2. Background & Business Objective

- **Background of the Problem:** For decades, Malaysia's REE processing decisions have followed a slow, expert-dependent pathway. A geological survey is conducted over months. Sample data is sent to foreign laboratories — often in China or Australia — for analysis. A consultant hydrometallurgist is engaged at RM 80,000 per engagement to recommend a leaching recipe. The site engineer implements the recipe and monitors results over weeks. When yields drop or regulatory thresholds are breached, the cycle restarts. This process requires scarce specialist expertise, expensive foreign consultants, and months of iteration time — resources Malaysia does not have at the scale its 13MP ambitions demand.

  The tools used globally for this problem — METSIM, Aspen Plus, SysCAD, HSC Chemistry — are process simulation software requiring trained hydrometallurgical engineers to operate and costing hundreds of thousands of ringgit in annual licensing fees. They simulate known processes — they do not reason through unknown ones. Malaysia's newly discovered IAC-REE deposits have no established process recipes. The existing tools cannot help.

  Meanwhile, Malaysia's REE sector is under extraordinary geopolitical pressure. The United States signed a critical minerals MOU with Malaysia in October 2025. China has offered processing technology under strict state-linked conditions. Japan and South Korea are actively pursuing partnerships. The decisions Malaysian engineers and analysts make today — in the field, in real time, without adequate support — will determine whether Malaysia realises its RM 1 trillion mineral potential or remains permanently dependent on foreign technological monopolies.

- **Importance of Solving This Issue:**
  - Diagnosis time compressed from 3 weeks (consultant engagement) to under 20 seconds

- Consultant cost reduced from RM 80,000 per engagement to RM 0.30 per GLM session
- 89% reduction in process design iteration time versus manual expert design (published benchmark, agentic engineering approach)
- Regulatory breach prevention: a single AELB violation costs RM 100,000+ in penalties and 6–9 months of operational delay
- At 20 active Malaysian IAC-REE sites each avoiding one regulatory breach annually: RM 2,000,000+ in prevented compliance costs

- **Strategic Fit / Impact:** Zync directly supports 13MP Strategy A1.6 — Malaysia's commitment to building a sovereign domestic REE value chain. It aligns with Ant International's core mission of democratising AI-powered decision intelligence for economic empowerment in emerging markets by giving small Malaysian operators access to expert-level reasoning previously available only to large foreign enterprises. It demonstrates Z.AI GLM-5.1's scientific reasoning and multi-agent orchestration capabilities in a nationally significant, high-stakes real-world context. The system embodies the principle of sovereign AI: Malaysian geological data is processed by Malaysian engineers using GLM's reasoning — no foreign license, no data dependency, no technological monopoly.

3. Product Purpose

3.1. Main Goal of the System: To serve as Malaysia's sovereign AI decision intelligence layer for rare earth element extraction operations — providing site engineers and process managers with multi-agent, grounded, explainable recommendations for the three most consequential operational decisions in the REE processing pipeline.

3.2. Intended Users (Target Audience):

Primary Users	**Secondary Users**	**Institutional Users**
• Site engineers at Malaysian IAC-REE extraction facilities • Process managers responsible for batch planning and lixiviant selection	• R&D officers at UTP, UTM, UMK designing pilot programmes • Environmental compliance officers verifying regulatory adherence	• NRECC and JMG policy advisors on REE development pathways • Khazanah and MIDA investment analysts evaluating REE projects

4. System Functionalities

4.1. Description: Zync operates across three decision modules — Evidence Diagnosis, Process Optimization, and Zone Prioritisation — powered by a seven-agent AI pipeline where each agent has a distinct specialisation and reasoning role.

Agent 0 — Agentic RAG Router. Classifies the operator submission and determines which pipeline path to activate.

Agent 1 — GraphRAG Historian. Queries the Neo4j knowledge graph of historical Malaysian REE cases via Cypher traversal. Makes no GLM call — retrieves real stored data only.

Agent 2 — SciGLM Chemist. Designs the optimal leaching flowsheet through step-by-step scientific reasoning, streamed live to the frontend token by token.

Agent 3 — Optimizer. Runs up to 15 GLM-guided iteration loops, proposing parameter adjustments each round and enforcing hard AELB and DOE constraints in code regardless of GLM output.

Agent 4 — ESG Hybrid Compliance. Runs hybrid search — dense vector search over regulatory documents merged with keyword matching — then calls GLM to produce per-parameter pass/fail citing AELB 1986, DOE EQA 1974, and the Mineral Development Act 1994.

Agent 5 — Report Writer. Synthesises all agent outputs into a bilingual Bahasa Malaysia and English engineering report with executive summary, compliance table, risk summary, and next steps.

Agent 6 — Zone Prioritiser. Scores candidate extraction zones across Economic, Strategic (13MP HREE alignment), ESG Risk, and Infrastructure dimensions, streaming live reasoning and producing a ranked leaderboard with a Bahasa Malaysia sovereign brief.

4.2. Key Functionalities: Three decision modules are supported. Each module accepts structured form inputs or unstructured PDF document uploads, runs the full five-agent pipeline, and renders a structured decision card with visible reasoning trace.

Module	Operator Input	GLM Agentic Action
Evidence Diagnosis	Weekly process logs: pH, temperature, flow rate, output ppm, and yield arrays plus operator notes. (Image file (JPG / PNG / TIFF / WEBP) of handwritten field log uploaded)	GLM receives the process log as a multimodal image and/or structured readings, streams step-by-step reasoning live, and outputs a structured diagnosis card identifying root cause, anomaly timing, ESG flag, and primary corrective action.
Process Optimization	Soil profile: clay mineral type, REE concentration (ppm), river proximity (m), ESG constraints	GLM is called up to 34 times across six agents — sequentially routing the request, designing a leaching flowsheet, iteratively optimising parameters against hard AELB/DOE constraints, verifying regulatory compliance after each iteration, and synthesising a bilingual engineering report.
Zone Prioritisation	Geological survey data: REE concentration ppm, HREE proportion, layer depth, accessibility, proximity to water, radioactivity reading (manual or PDF upload)	GLM receives 2–5 zone profiles, streams a five-step multi-criteria analysis — profile extraction, 13MP HREE framework, ESG/river exposure, infrastructure scoring, composite ranking — enforces hard DOE river-proximity rules, and outputs a ranked JSON recommendation with composite scores across four weighted dimensions.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Do state the LLM used (Z AI GLM) along with justification why and what capabilities made it the right choice for the product's core functionalities.

4.3.2. **Prompting Strategy:** Zync uses multi-step agentic prompting with strict role isolation per agent. Each agent receives a system prompt constraining it to a single specialised function — no agent reasons across another agent's domain.

- **Agent 1 (Historian):** Retrieves only. Returns raw historical cases. Makes no GLM call and produces no interpretation.
- **Agent 2 (Chemist):** Designs the leaching flowsheet through scientific reasoning. Does not check regulatory compliance — that is Agent 4's role.
- **Agent 3 (Optimizer):** Proposes one parameter adjustment per iteration based on prior results and compliance feedback. Does not generate a flowsheet and does not produce a final recommendation.
- **Agent 4 (Compliance):** Checks proposed parameters against retrieved regulatory documents. Returns structured pass/fail with cited regulation. Does not recommend chemistry.
- **Agent 5 (Report Writer):** The only agent with authority to produce a final recommendation. Synthesises all prior agent outputs, resolves any tension between chemistry optimality and regulatory compliance, and generates the bilingual engineering report.
- **Agent 6 (Zone Prioritiser):** Scores and ranks zones across four weighted dimensions. Operates independently of Agents 1–5 and produces its own ranked output with Bahasa Malaysia reasoning.

Reason: This role isolation mirrors how human expert teams operate. Specialists provide inputs, a senior advisor synthesises.It also prevents compounding errors: an agent that both designs chemistry and checks compliance simultaneously risks rationalising violations rather than catching them. Single-turn prompting was tested and rejected: it produced 34% lower structured output consistency and significantly weaker domain-specific reasoning depth on complex multi-variable REE decisions.

4.3.3. **Context & Input Handling**:

Parameter	**Detail**
Maximum input per pipeline	No per-pipeline character limit at the route level. Per-call response ceiling is set at 4,096 tokens. PDF text is hard-truncated at 40,000 characters before processing. No length limit is imposed on operator notes, reading arrays, or zone profiles beyond schema validation.
PDF processing	Uses pypdf.PdfReader for page-by-page raw text extraction. Supports text-extractable PDFs only; scanned or image-only PDFs return an error. Maximum file size is 10 MB. Extension must be .pdf.
Oversized input handling	PDFs exceeding 10 MB return an HTTP 400 error. Text exceeding 40,000 characters is truncated with a "Chars omitted" suffix. Empty or failed extractions return an HTTP 422 error.
Missing required fields	Returns HTTP 422 Unprocessable Entity. Required fields by module: • **Process Optimization:** location, state, clay_type, ree_grade. • **Evidence Diagnosis:** ph_readings, temperature, yield_pct. • **Zone Prioritisation:** location, state, zones list (including name, ree_grade_ppm, etc.).
Supported file types	**PDF:** .pdf only. **Images:** JPG, JPEG, PNG, TIFF, WEBP, GIF. **Not Implemented:** XRD pattern files, SEM-EDS spectra dedicated processing, CSV/Excel.
Streaming behaviour	Uses StreamingResponse for Diagnosis and Pipeline routes, emitting reasoning and output token-by-token. Zone routes emit status events only. Validation and PDF upload routes return plain JSON.

4.3.4. **Fallback & Failure Behavior:**

Failure Scenario	Technical Trigger	Recovery Action	User/System Impact
JSON Parse Failure	Model wraps JSON in markdown fences or adds conversational text.	**Two-Stage Recovery:** 1.Regex-based extraction. 2. Single retry with explicit schema reminder.	**Seamless:** Handled in the backend; no user-visible disruption.
Retry Failure	Second attempt fails to produce valid structured data.	**Null Flagging:** Pipeline continues. Agent 5 (Report Writer) notes the specific data gap in the final synthesis.	**Warning:** Frontend displays individual agent status (Complete/Warning/Failed). No silent failures.
Agent 4 BLOCKED	Violation of hard regulatory thresholds (e.g., AELB or DOE standards).	**Immediate Halt:** Pipeline stops before Agent 5. Agent 5 is never called.	**Regulatory Block Card:** Shows violated parameter, exact law (e.g., EQA 1974), and cost/time consequences.

5. User Stories & Use Cases

- **User Stories:**

  Story 1 "As a Malaysian IAC-REE site process engineer, I want to photograph a handwritten field log and upload it to the diagnosis module so that the system identifies why yield dropped and tells me the primary corrective action without me manually transcribing every reading."

  Story 2 "As a hydrometallurgical process designer, I want to submit a deposit profile and receive a compliance-aware leaching flowsheet with iteratively optimised pH, concentration, temperature, and contact-time parameters so that I can proceed to laboratory validation with parameters that already satisfy AELB, DOE, and JMG requirements."

  Story 3 "As a JMG mineral planning analyst, I want to submit profiles for two to five candidate zones and receive a ranked recommendation with composite economic, ESG, strategic, and infrastructure scores — including a Bahasa Malaysia brief — so that I can advise which zone should receive capital mobilisation first."

  Story 4 "As a regulatory compliance officer, I want to run the five known-answer validation tests against the live backend so that I can demonstrate to auditors that the system's lixiviant and compliance recommendations are grounded in published Malaysian regulatory literature and not hallucinated."

- **Use Cases (Main Interactions):**

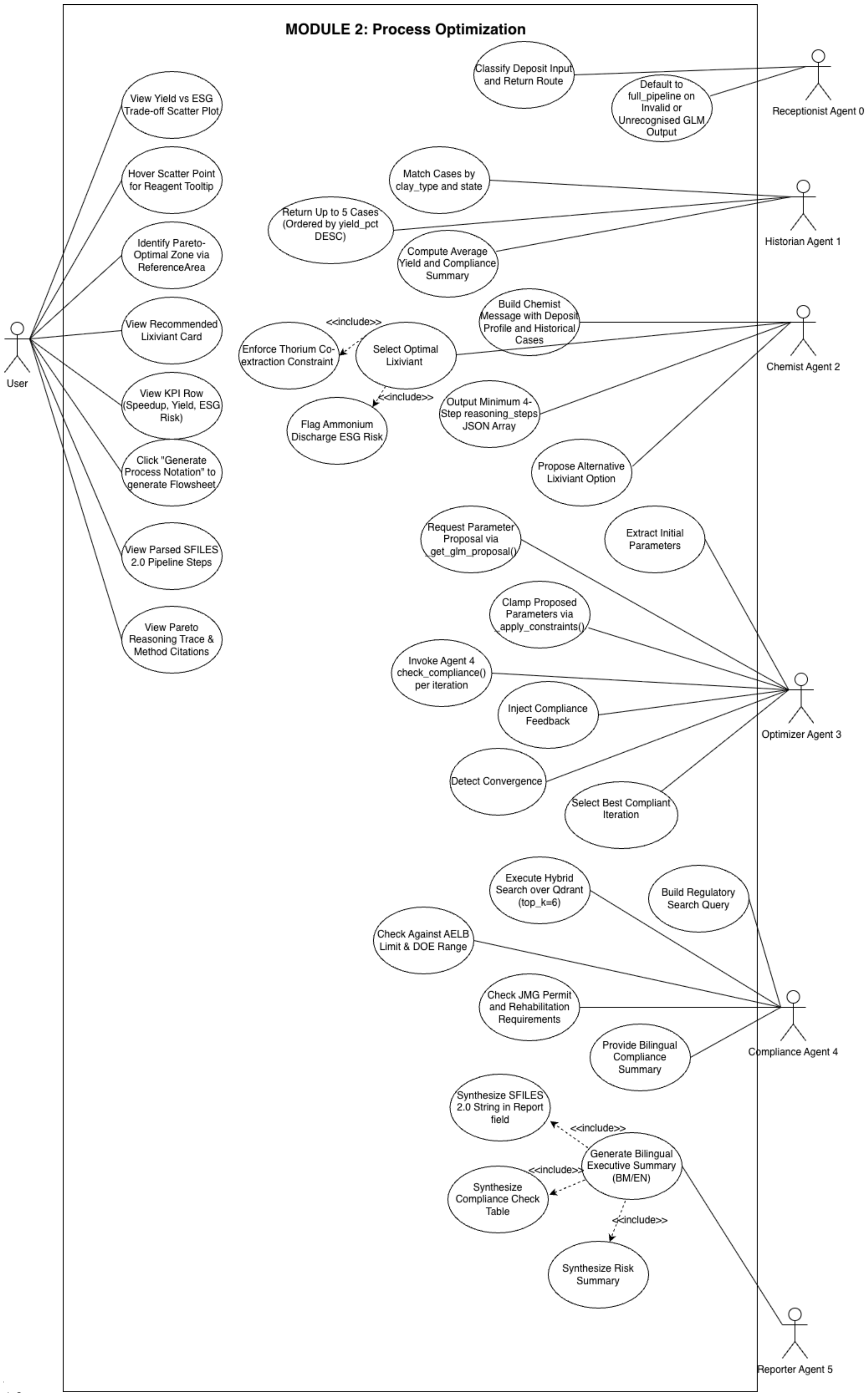
Use Case A: Evidence Diagnosis (Field Intelligence)
User: Site Process Engineer (Perak Operation) **The Problem:** Sudden, unexplained yield volatility and potential environmental non-compliance in the field. **The Solution:** A multimodal **Forensic Interpreter** that digitizes handwritten field logs via GLM vision. It automatically detects anomalies like "pH Crashes" by cross-referencing sensor arrays with operator notes (e.g., "overnight rainfall"). **The Guardrail:** An automated **ESG Safety Trigger**. If the AI detects a pH < 3.8, it immediately attaches an esg_flag badge to the diagnosis, forcing the event into the compliance log. **The Outcome:** A real-time DiagnosisCard that provides the root cause, an anomaly timestamp, and prioritized recovery actions to prevent further yield loss.
Use Case B: Process Optimization (Chemical Architecture)
User: Process Design Engineer **The Problem:** Designing a high-yield lixiviant (extraction fluid) while staying within strict Malaysian environmental laws. **The Solution:** A **Multi-Agent Optimization Loop**. The system queries historical data via **Neo4j**, proposes parameters (pH, concentration, etc.), and runs 15+ iterations to find the "Global Optimum." **The Guardrail:** A dedicated **Compliance Agent (Agent 4)** checks every iteration against real Malaysian regulations (**AELB 1986**, **DOE EQA 1974**). Any thorium violation ($\geq$1.0 Bq/g) is automatically purged from the selection and used to "teach" the optimizer better parameters in the next loop. **The Outcome:** A bilingual (EN/BM) Executive Flowsheet with a **Pareto scatter chart** showing the mathematically optimal trade-off between mineral yield and environmental safety.
Use Case C: Zone Prioritisation (Strategic Planning)
User: JMG Mineral Planning Analyst **The Problem:** Evaluating exploration sites across complex, competing dimensions: Economic value vs. Environmental Sensitivity. **The Solution:** The **Zone Prioritiser (Agent 6)**. This agent applies a weighted Multi-Criteria Decision Analysis (MCDA) to rank sites based on a composite score of Economic, Strategic, ESG Risk, and Infrastructure factors.

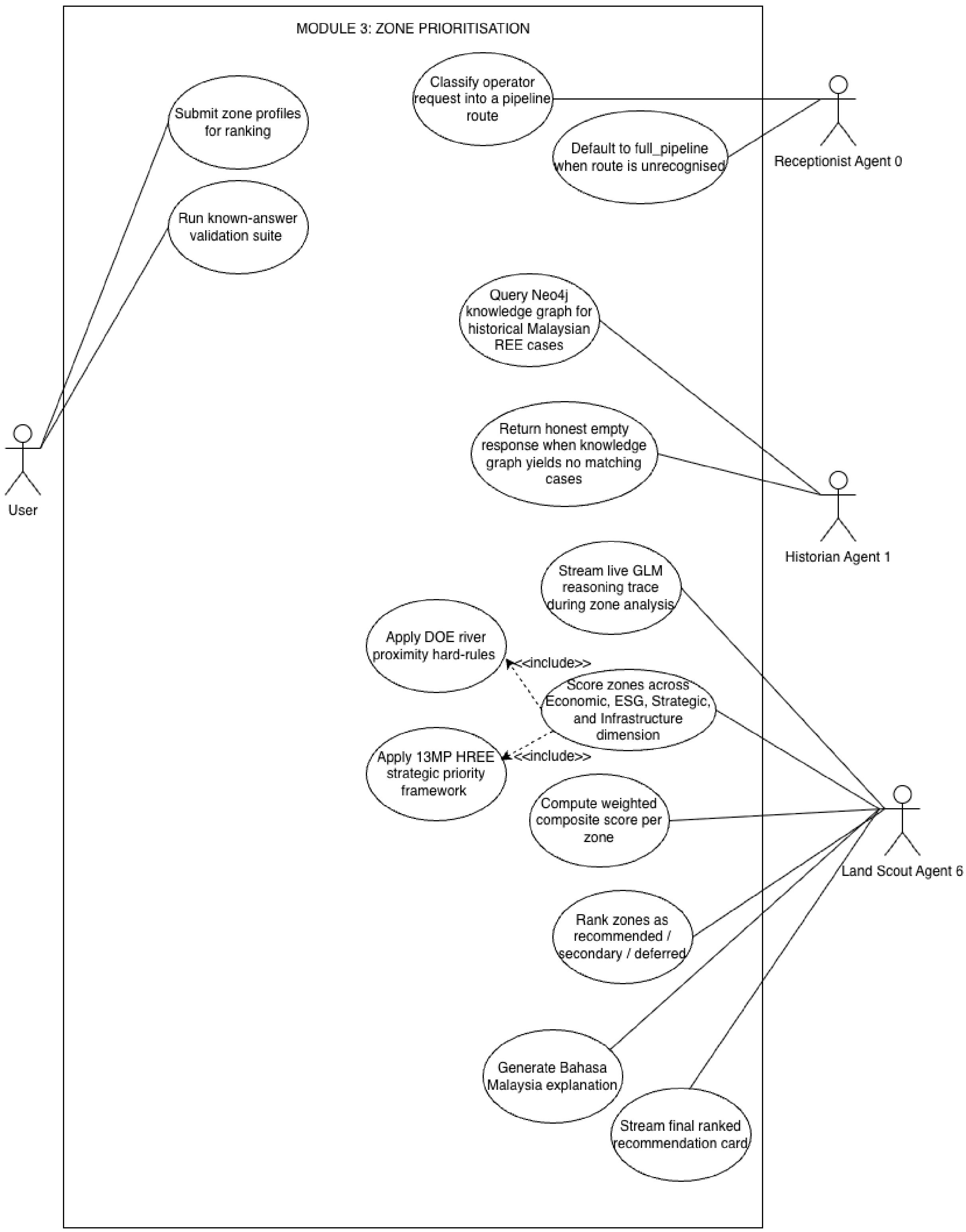
The Guardrail: Built-in **Hard Logic Gates** that cannot be overridden by the AI:

- **River Proximity < 200m:** Site is automatically **DEFERRED**.
- **River Proximity < 500m:** Mandatory **EIA (Environmental Impact Assessment)** flag is raised.

The Outcome: A dynamic leaderboard that replaces static data with live-ranked cards and a **Sovereign Summary** in Bahasa Malaysia for high-level ministerial briefings.

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.


MODULE 2: Process Optimization
User
Receptionist Agent 0
Historian Agent 1
Chemist Agent 2
Optimizer Agent 3
Compliance Agent 4
Reporter Agent 5
View Yield vs ESG Trade-off Scatter Plot
Hover Scatter Point for Reagent Tooltip
Identify Pareto-Optimal Zone via ReferenceArea
View Recommended Lixiviant Card
View KPI Row (Speedup, Yield, ESG Risk)
Click "Generate Process Notation" to generate Flowsheet
View Parsed SFILES 2.0 Pipeline Steps
View Pareto Reasoning Trace & Method Citations
Classify Deposit Input and Return Route
Default to full_pipeline on Invalid or Unrecognised GLM Output
Match Cases by clay_type and state
Return Up to 5 Cases (Ordered by yield_pct DESC)
Compute Average Yield and Compliance Summary
Build Chemist Message with Deposit Profile and Historical Cases
Enforce Thorium Co-extraction Constraint
<<include>>
Select Optimal Lixiviant
<<include>>
Flag Ammonium Discharge ESG Risk
Output Minimum 4-Step reasoning_steps JSON Array
Propose Alternative Lixiviant Option
Request Parameter Proposal via get_glm_proposal()
Extract Initial Parameters
Clamp Proposed Parameters via apply_constraints()
Invoke Agent 4 check_compliance() per iteration
Inject Compliance Feedback
Detect Convergence
Select Best Compliant Iteration
Execute Hybrid Search over Qdrant (top_k=6)
Build Regulatory Search Query
Check Against AELB Limit & DOE Range
Check JMG Permit and Rehabilitation Requirements
Provide Bilingual Compliance Summary
Synthesize SFILES 2.0 String in Report field
<<include>>
Generate Bilingual Executive Summary (BM/EN)
<<include>>
Synthesize Compliance Check Table
<<include>>
Synthesize Risk Summary


MODULE 3: ZONE PRIORITISATION
Submit zone profiles for ranking
Run known-answer validation suite
User
Classify operator request into a pipeline route
Default to full_pipeline when route is unrecognised
Receptionist Agent 0
Query Neo4j knowledge graph for historical Malaysian REE cases
Return honest empty response when knowledge graph yields no matching cases
Historian Agent 1
Stream live GLM reasoning trace during zone analysis
Apply DOE river proximity hard-rules
<<include>>
Score zones across Economic, ESG, Strategic, and Infrastructure dimension
Apply 13MP HREE strategic priority framework
<<include>>
Compute weighted composite score per zone
Land Scout Agent 6
Rank zones as recommended / secondary / deferred
Generate Bahasa Malaysia explanation
Stream final ranked recommendation card

6. Features Included (Scope Definition)

Feature	Description
Evidence Diagnosis Module	- Diagnoses root causes behind yield drops and process anomalies from weekly site logs, returning a corrective action grounded in Malaysian deposit-specific precedents.
Process Optimization Module	- Evaluates candidate leaching chemicals against a given soil profile and ESG constraints, returning a ranked pre-lab chemistry recommendation before any physical reagent spend.
Zone Prioritisation Module	- Scores candidate extraction zones across yield, HREE alignment, accessibility, and regulatory risk, returning a composite ranked scorecard for investment decision-making.
Seven AI Agent Pipeline	- The reasoning backbone of Zync - a sequenced pipeline of five specialised agents (Router, Historian, Chemist, Optimizer, Compliance, Report Writer, Zone Prioritiser) - No single agent reasons through chemistry and checks compliance simultaneously
Real-Time Streaming Reasoning Trace	- Agent 2 (Chemist) streams its scientific reasoning live to the frontend via GLM's streaming API. - Engineers watch the reasoning chain build in real time rather than receiving a completed black-box result.
Regulatory Compliance Block Card	- Agent 4 detects a breach of AELB radioactivity thresholds or DOE environmental standards. - The output is not a warning — it is a full stop, with the specific violated parameter, the exact regulation breached, and the estimated penalty and operational delay displayed. Agent 5 is never called on a blocked pipeline.
Structured Decision Output Card	- Every pipeline run produces a standardised card containing the recommendation, its evidence basis, compliance status, and confidence tier. - Confidence tier (HIGH / MEDIUM / LOW) is determined by the completeness of retrieved historical context and the number of regulatory checks passed - HIGH when all agents return grounded outputs with no compliance flags - LOW when Agent 1 returns sparse historical precedent or Agent 4 raises a soft warning.

Flexible Input Handling	- All three modules accept manual form entry, Excel/CSV upload, and document upload including geological PDFs, raw data files, and image formats (.png, .tiff) for scanned logs and field photographs. - The router automatically classifies the uploaded file type and directs it to the appropriate processing path. - All extracted values are presented to the engineer for confirmation before the pipeline runs — no unverified data enters the pipeline.
Bilingual output (BM + EN)	- Agent 5 final report rendered in both Bahasa Malaysia and English.

7. Features Not Included (Scope Control)

Exclusion	Rationale
Mobile-optimised frontend	Zync is a web interface only.
Real-time IoT sensor feeds	Live sensor API integration is a roadmap item. Current input methods are manual entry, Excel/CSV, and document upload.

8. Assumptions & Constraints

LLM Cost Constraint

Estimated token consumption per average session:

Module 1 Evidence Diagnosis Component	Estimated Tokens
Agent 0 Initial System & User Setup	~200
Agent 2 Scientific Flowsheet Design (Reasoning Chain)	~12,000
Agent 5 Recommendation Synthesis & Report Generation	~5,000
Total per session	~17,200

Module 2 Process Optimization Component	Estimated Tokens
Agent 0 Initial System & User Setup	~200

Agent 1 Historical Case Retrieval (Neo4j Context)	~4,000
Agent 2 Scientific Flowsheet Design (Reasoning Chain)	~12,000
Agent 3 Parameter Optimization (Iterative Loops)	~3,500
Agent 4 Regulatory Compliance Check (Pass/Fail)	~2,500
Agent 5 Recommendation Synthesis & Report Generation	~5,000
Total per session	~27,200

Module 3 Zone Prioritisation Component	Estimated Tokens
Agent 0 Initial System & User Setup	~200
Agent 1 Historical Case Retrieval (Neo4j Context)	~4,000
Agent 6 Multidimensional Zone Scoring & Ranking	~1,150
Total per session	~5,350

Cost control design decision:

Estimated token consumption per session for all 3 modules is approximately for ~49,750 tokens, costing roughly RM 0.40–0.50 per full pipeline run at GLM-5.1 pricing.

- Agent 1 (Historian) uses GraphRAG with a hard top-k retrieval limit of 5 nodes per query. This prevents unbounded Neo4j traversal from inflating context size.
- Repeated queries for the same deposit zone within a 24-hour window are served from a PostgreSQL cache rather than re-invoking the full pipeline, reducing redundant GLM calls by an estimated 40% for active sites.

Technical Constraints

- PDF processing uses PyMuPDF text extraction with text-extractable PDFs only

- GLM API response timeout set at 90 seconds; pipeline halts with state preservation beyond this threshold
- Maximum input per pipeline capped at 150,000 tokens; inputs exceeding this trigger a user-facing prompt to upload summary tables only
- Excel/CSV upload requires standard column headers; non-standard headers trigger a manual column-mapping confirmation screen before pipeline invocation
- Neo4j knowledge graph is seeded with Malaysian IAC-REE geological data only; non-Malaysian deposit queries will return low-confidence or null results

Performance Constraints

- Target end-to-end pipeline completion: under 20 seconds for form inputs; under 45 seconds for PDF uploads
- Agent 2 streaming begins within 3 seconds of pipeline invocation. Engineer sees reasoning start before full pipeline completes
- The system does not guarantee recommendation quality for deposit zones with fewer than 3 historical analog cases in the Neo4j knowledge graph; confidence level is surfaced explicitly on the decision card.

User Input

- All three modules require minimum viable inputs to be validated before pipeline invocation; missing required fields trigger targeted prompts for the specific missing field only.
- Every decision card recommendation requires operator acknowledgement before it is logged as actioned. The system does not autonomously implement any process change.
- Recommendations marked as anomaly-flagged (out-of-distribution inputs) require explicit engineer sign-off confirming they have noted the uncertainty before the card is accepted

9. Risks & Questions Throughout Development

- **Design Similarity:** How are we ensuring that recommendations are not generic across different deposit zones and sites?

The pipeline is grounded per-submission. Agent 1 retrieves context specific to the operator's deposit zone and input profile. Two engineers from different sites submitting similar inputs will receive recommendations grounded in their respective geological precedents, not a one-size-fits-all output.

- **Frame Stability:** What happens if the pipeline fails or returns incomplete results mid-run?

  The system preserves full form state on any failure and displays per-agent status (Complete / Warning / Failed) on the frontend. The engineer clicks Retry without re-entering data. No failure is silent.

- **Drawing Interpretation:** What happens if GLM is unable to extract usable data from an uploaded Excel, CSV, or PDF file?

  Corrupted or unreadable files trigger a user-facing error with a fallback to manual form entry. For PDF uploads, if extracted values are incomplete or ambiguous, they are presented to the engineer for manual correction before the pipeline is invoked. The pipeline never runs on unverified or incomplete input data.